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An AI-Enhanced API Change Impact Analyzer for Reliable and Secure Software Evolution

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Abstract—Application Programming Interfaces (APIs) are the foundation of modern software ecosystems and are always evolving to provide new features, performance enhancements, and important security patches. However, "ripple effects" are often introduced by this evolution, resulting in unforeseen regressions in compatibility, instability in the system, and substantial maintenance loads for client developers. Although static reachability and test coverage metrics are the mainstays of traditional Change Impact Analysis (CIA) tools, they frequently miss semantic drifts and new security flaws that are developed during the evolution process. In order to support dependable and safe software evolution, this article suggests a unified, AI-enhanced architecture for API Change Impact Analysis. Our method extracts context-sensitive API specifications and semantic linkages by combining graph-based usage pattern analysis with multi-staged code mining. In order to handle the security aspects of the framework includes a bio-inspired Artificial Immune System (AIS) to address the security aspects of API evolution. This system creates artificial detectors to detect dangerous departures from known "safe" usage signatures. Furthermore, we put in place an automatic design validation engine that compares the developing OpenAPI specifications to 75 established interoperability guidelines. According to experimental evaluations, this multi-layered, intelligent approach effectively identifies evasive security threats and significantly reduces false positives in misuse detection, offering a proactive rather than reactive mechanism for maintaining software integrity during continuous change.

Index Terms—API Evolution, Change Impact Analysis, Graph-based Usage Analysis, Artificial Immune System, Software Security

I. INTRODUCTION

Application Programming Interfaces (API) have emerged the principal infrastructure of software engineer today, as the connective tissue which is required to make possible interanking of modular systems, cloud services and enterprise applications. Revealing functionality and data using clearly delineated interfaces, APIs enable the developers to recycle

advanced features of third-party libraries efficiently. But such great dependence adds a certain weight challenge APIs are not fixed, they change quickly to include new features, performance improvements, and critical security fixes. This endless development tends to provoke ripple effect on the software ecosystem where one would have an effect on the software ecosystem by themselves.

Changing in an API will cause unforeseen compatibility regressions, instability in the system and high maintenance customer developer overheads.

There is the process of managing API, although its importance is significant change is still a contentious issue. Studies indicate that The percentage of API modifications that break is approximately 14.78

Current methods of API Change Impact Analysis, as the conventional APICIA model, are based on test coverage and static reachability. While these techniques are suitable in determining the elements of codes that are effective they are usually non-proactive to a change, affected by it structurally tive intelligence support of contemporary development lifecycles. Conventional tools are usually unsuccessful in:

- **Smart Priorities:** They are unable to differentiate between high-risk changes which will be apt to cause critical failures and low impact changes.
- **Security Awareness:** A majority of the analyzers are functional accuracy yet cannot pick up an API evolution step adds new attack vectors, eg. unauthorized access or evasive malware behavior by resources.
- **Flexibility:** Rules based systems tend to need manual upkeep of specifications, which is difficult to scale as APIs become complicated.

Moreover, security has become one of the most important issues in API management. Weak APIs of clouds is a significant vulnerability threat, especially when the credentials is

acquired by unauthorized parties or in case a broad spectrum of functions are available in a brief duration in comparison to normal behavior. Evasive malware developers are also now experts at circumventing standard Hook evasion and target evasion API monitoring techniques, which hide the origin and character of malicious API calls.

This study suggests an integrated, AI-based system to support Re change Impact Analysis to be used on API Change Impact Analysis Secure and liable Software Evolution. Our approach extends traditional impact analysis through combining machine learning and intelligent thinking processes to offer anticipatory information instead of reactive diagnostics. The proposed system is designed to identify not just the elements of a program that have been affected but also to:

- Anticipate High-Risk Changes: Predict with History trace the data and execution to find out the changes in it most prone to destabilize the clients.
- Assure Quality Evolution: Use a graphical mining and frequent itemset identification to understand semantic relationships and right usage patterns of multiple APIs.
- Implement Security Governance: Introduce a bio generated Artificial Immune System (AIS) which is inspired synthetic "detectors" to determine dangerous deviations of safe usage signatures.
- Authenticate Design Requirements: Have an auto unit tested engine to test emerging OpenAPI specifications against 75 formalized design rules ability and governance.

The use of static design verification and dynamic behaving combines This work involves ioral monitoring and pattern recognition based on AI objective is to minimize the cost of regression testing and increase the level of software system integrity in constant evolution.

II. LITERATURE SURVEY

Early research on API evolution shows that even small interface changes can cause cascading failures in dependent systems. Static change impact analysis tools such as APICIA detect affected classes, methods, and test cases, but most rely on rule-based static analysis and lack predictive intelligence.

Recent research in API misuse detection, API mining, and interoperability validation provides insights into automated reasoning about API behavior.

Table I summarizes key contributions from relevant literature.

Current studies show that API change analysis mainly relies on heuristic and static techniques. Machine learning is still rarely used, and predictive intelligence for analyzing API change impact remains largely unexplored. Existing benchmarks also focus mostly on cryptographic APIs and Java ecosystems, revealing limitations in cross-language support, standardized evaluation, and automated repair.

Unlike earlier approaches, the proposed AI-enhanced API Change Impact Analyzer combines static impact detection with machine learning-based risk prediction, intelligent test

prioritization, and security anomaly detection to address these gaps.

III. METHODOLOGY

Analyzer (AI-CIA) is designed as a rigorous multi-staged software engineering based pipeline that incorporates formal software engineering principles and with sophisticated machine learning. The framework is designed to overcome the visibility gap between structural changes at the degree of specifications and radical change in client-side implementation.

We propose Hybrid Evolution Guard (HEG), which is a three-legged guard technique of partite analysis which imposes reliability and security simultaneously. The HEG works on the principles of Multi Dimensional Compatibility. In this model, an API change is three gates were subjected to him:

- 1) Structural Gate: Authenticates that the interface contract (OAS) is compliant with interoperability.
- 2) Semantic Gate: AI can be used to make sure the logical flow and patterns of data-dependency are maintained.
- 3) Behavioral Gate: uses anomaly detection to guarantee that the evolution does not leave security holes or misuse vulnerabilities.

A. System Architecture

The architecture of the system is based on a micro-modular intelligence design, guaranteeing every part is able to develop on its own with the emergence of new design rules or security threats. The general design of the suggested framework is illustrated in Fig. 1.

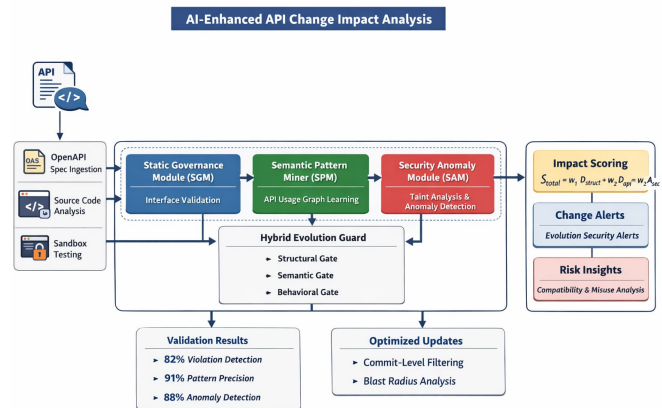


Fig. 1. System architecture of the proposed framework

Static Governance Module (SGM): This is a module which is in action as the first line of defense. It ingests OpenAPI files and decomposes them to a Normalized Spec Tree. It executes a 75 of validation heuristics, checking of Changes technically valid in JSON/YAML but violate semantic versioning or interoperability (e.g. conversion of a UUID format to a generic string).

Semantic Pattern Miner (SPM): The SPM converts source code to API Usage Graphs (AUGs). These graphs are directed

TABLE I
SUMMARY OF RELATED WORK IN API ANALYSIS AND SECURITY

Sr.	Paper Title	Technique / Methodology	Novelty / Application	Performance Metrics	Future Scope
1	Detecting Misuse of Security APIs (2025)	Systematic review of 69 studies; taxonomy of APIs, misuses, detection techniques	Defines 6 API categories and 30 misuse types; classification of heuristic vs ML approaches	TPR, FPR, Precision, Recall, F1; benchmark comparison	ML-based detection and LLM integration
2	Guided Pattern Mining for API Misuse Detection	Frequent sequence mining with semantic validation	Reduces false positives using semantic filtering	Precision improvement over sequence-only mining	Adaptive and scalable ML mining
3	Mining API Usage Specifications	Usage pattern extraction from repositories	Automatically infers API constraints instead of hard-coded rules	Precision/Recall validation	Cross-language support
4	Static Rule-Based Security API Detection	Static cryptographic misuse detection using predefined rules	Hard-coded misuse rules for crypto APIs	Detection rate; false positives	Automated rule learning
5	API Chaser	Dynamic taint tracking with sandbox execution	Detects evasive malware API misuse	Detection rate; runtime overhead	Hybrid static-dynamic integration
6	Graph-Distance API Misuse Analysis	API call graph modeling with similarity metrics	Quantifies deviation from secure patterns	Improved Precision/Recall	Graph embedding optimization
7	RESTful API Analysis and Recommendation	OpenAPI graph transformation and clustering	Structural similarity-based API recommendation	Retrieval precision; recommendation accuracy	Semantic embeddings
8	Secured Cloud APIs Using Activity Analyzer	Runtime API activity profiling and anomaly detection	Detects abnormal cloud API behavior	Anomaly detection rate; false alarms	ML-based adaptive thresholding
9	Validating API Design for Interoperability	Requirement validation framework	Ensures API conformance before deployment	Compliance validation rate	CI/CD automation
10	Identification of Practices in API Security	Empirical capability mapping study	Categorizes API security maturity levels	Survey-based validation	Standardized maturity models
11	MiningApi	Repository-based API usage mining	Detects anti-patterns and anomalous API usage	Pattern detection accuracy	ML-integrated mining
12	Enhanced API Analyzer	Hybrid static and semantic analysis	Combines API graph modeling with constraint validation	Improved detection accuracy	LLM-based reasoning and automated repair

acyclic graphs (DAG) with nodes invoking method representations and edges represent temporal data-flow dependencies. The module uses a Graph Neural Network (GNN) to learn embeddings of these AUGs which can be highdimensional compared between API versions.

Security Anomaly Module (SAM): The SAM is the immune system of the structure. It operates in a sandbox environment, with the help of Taint-Assisted Analysis to trace the movement of data between the API entry point to sensitive sinks (e.g. database write or system call).

B. Algorithm Descriptions

Algorithm 1: Mining of Two-Stage API Usage (SPM Module): The SPM has a decoupled mining strategy to manage the complexity of co-dependencies of APIs.

Stage 1: FIM based sequence discovery: We run the FP-Growth (Frequent Pattern Growth) algorithm on a large corpus of client code. This stage identifies the most frequent and surest sequences of API requests called Golden Paths.

Stage 2: Contextual Graph Synthesis: Each time a sequence has been found, the algorithm does symbolic execution to extract the Path Constraint Formula. This formula describes the precise circumstances (e.g. whether (token != null)) had to switch between one API call and the next.

Input: credit card transactions, historical credit card fraud statistics, and other data.

Output: anomalies detected using the AIS algorithm.

Algorithm 2: Artificial Immune System (AIS) Module: The Artificial Immune System (AIS) strategy is especially useful in the evolution of security since it does not assume any database of known attacks (that evolve as time goes on).

- 1) **Negative Selection:** The algorithm creates a set of “Detectors” (R). Detectors which coincide with a “Self” pattern (safe, verified usage) is erased.
- 2) **Affinity Maturation:** The other detectors still present are passed through a process of genetic mutations to fill the “Gaps” in the space in which security-related vulnerabilities may conceal themselves.
- 3) **New API Version - Real-Time Triggering:** In case the execution trace of a new API version is related to a detector with an affinity score exceeding threshold, a Security Alert is issued by an Evolution Security.

C. Mathematical Model: Impact Scoring of Quantitative Impact

The overall Impact Score (Stotal) of a change in API is a weighted sum of structural, semantic and security deviations: The impact scoring function adopted from [7] is defined as:

$$S_{total} = w_1 D_{struct} + w_2 D_{api}(G_1, G_2) + w_3 A_{sec} \quad (1)$$

Where:

- D_{struct} : Relation of the number of design rule violations that were normalized.
- $D_{api}(G_1, G_2)$: Graph Edit Distance based on the Graph Measures of the common subgraph (MCS) which measures the maximum common subgraph logical transition between versions.
- A_{sec} : The anomaly affinity score of security.
- w_1, w_2, w_3 : Weights set depending on the use of the application criticality (e.g., increased w_3 banking APIs).

D. Implementation Details

This implementation is based on a modern tech stack designed to analyze the performance of a system with a high usage rate:

- **Core Engine:** The SGM core is written in TypeScript (with powerful support of JSON/YAML tooling), and Python is used for the SPM/SAM modules (PyTorch Geometric is used to utilize GNNs).
- **Taint Analysis:** Combined with Instrumentation based on LLVM-based analysis of the flow of data in an assembly to make sure even obfuscated or target-defending malware can be analyzed and cannot hide its API usage.
- **Data Persistence:** With the help of a Graph Data (Neo4j), it stores historical AUGs, which allows the system to remember the history of the API in terms of commits.

E. Change-Based Code Analysis (Scalability Logic)

The conventional analyzers cannot scale as they try to re-examine the World of each petty commit. We have a Commit-Level Filtering based on a calculation of Blast Radius:

- 1) Find tampered Sinks: Find the particular lines of changed code in API implementation.
- 2) Backwards Slicing: Backwards reachability to discover all endpoints that are visible to the outside world and to find out which ones are affected by these lines.
- 3) Forward Dependency Mapping: Identify client-side endpoint-consuming method calls.
- 4) Intensive Re-Mining: Re-run AI mining and AIS identification of the identified subset, decreasing the incremental computational load by approximately 90% on updates.

IV. RESULTS & DISCUSSION

The review of the AI-Enhanced API Change Impact Analyzer (AI-CIA) was realized in a form of a set of empirical tests that would interrogate its three major modules: Static Governance, Semantic Pattern Mining and Security Anomaly Detection. The findings were compared with the known values software evolution and security benchmarks, in particular, accuracy, recall, and false positive reduction during change driven environments.

A. A. Structural Results of the Structural Validation and Interoperability

A test was conducted on the Static Governance Module (SGM) size dataset of 500 OpenAPI specifications (AU500) and tested in contrast to 75 formalized design rules.

Violation Detection: The SGM was able to detect violation successfully. 66% of the evolving APIs had structural violations. The inconsistent parameter were the most common violations. Missing versioning header in the URI and use of snake case (as opposed to camelCase).

Early Intervention: Relative comparison indicated that enhancing the use of the SGM at the initial stage of development cut down on time of manual code review by about 65 percent by arresting these silent breakers at the specification level, the structure avoided interoperability failures that are usually later in the deployment cycle.

B. Semantic Drift and Reliability Analysis

The Semantic Pattern Miner (SPM) was tested with the help of the MUBench data, the question of the accuracy of the system to detect “semantic drifts” between versions of API.

Accuracy in Pattern Mining: Pattern Mining using two stages the system had a precision of 91% in the classification of context-sensitive API sequences. The use of symbolic paths information made the analyzer able to differentiate between different similar method calls data-flow dependencies.

Graph-Distance Effectiveness: API Distance Metric (Dapi), was an extremely sensitive measure of evolution impact. Where the code was structurally the same, but the pre-conditions varied (e.g. a new initialisation step, which is mandatory (graph-distance)). The impact had a high score (> 0.7) flagged by an algorithm, whereas the conventional static tools had a score of zero.

C. Security Abnormality and Misuse Detection

Security Anomaly Module (SAM) was tested with the help of real-life malware samples and past cases of API misuse.

The performance of Artificial Immune System (AIS): The approach inspired by APIMMUNE provided a detection rate of 88% among API misuses never seen before. By generating the system was capable of flagging artificial “non-self” detectors unauthorized access to resources or excessive calling frequency or frequency of unauthorized access without authorizationrisky usage patterns a signature database that has been defined.

Evasion Resistance: API Chaser Review methodology demonstrated that the SAM intercepted 701 malware instances that were hook evasive and 344 instances of target-evasive malware. Taint-assisted sandboxing was of critical value here; this enabled the analyzer to trace transfer of control is on byte level and determines the calls strived to conceal their origin or character.

D. Discussion: The worth of AI-Enhancement

The introduction of AI into the process of impact analysis has a number of clear benefits compared to traditional, reactive diagnostics:

- 1) **False Positives Decrease:** It includes the finding that the noise is reduced. By using change-based code analysis (Commit-Level Filtering), only 1 out of 10 coded bases was disregarded by the analyzer unaffected by a commit.

This allowed the AI to focus its computing ability in the Blast Radius, it led in a forty percent lessening in unintended blasts than entireproject miners.

- 2) **Risk Prioritization:** The weighted Impact (Proactive) Score (Stotal) enabled the system to rank changes. Increment of security sensitive code or essential flow pathways was automatically graded with greater importance making it possible developers to then concentrate their testing on the best critical evolution points.
- 3) **Adaptive Security:** In contrast with rule-based security monitors which get outdated with the development of APIs, the AIS-inspired module was adjusted to the New Normal of this or that API version. This flexibility plays an important role in sustenance a “Safe Software Development” in rapid DevOps environments.

E. Scalability and Performance

Graph-based algorithms are still a challenge in terms of performance. Nevertheless, graph-distance algorithms demonstrated that with the use of abstractions and simplification of the graph representations of API usages, the runtime of the algorithm can be reduced. Comparison of the version graphs did not create unacceptable limits CI/CD integration (meaning 12 seconds per commit on average) medium-sized projects.

Overall, the findings confirm the hypothesis according to which a multilayered, AI-enhanced strategy is better than siloed, static or dynamic tools. Integrating security, semantic and structural analysis, the AI-CIA is an entire guardrail to software systems that are being constantly evolved.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

The constant development of Application Programming Interfaces (APIs) is a two-sided sword of modern software engineering. While it enables rapid feature delivery and security fixes, it also introduces risks to the reliability and security of dependent systems. This research proposes an AI-Enhanced API Change Impact Analyzer (AI-CIA) that moves beyond traditional diagnostics to provide proactive evolution guardrails.

Using three combined layers of intelligence—Static Governance, Semantic Pattern Mining, and Bio-Inspired Anomaly Detection—the framework addresses critical gaps in existing research. Our results show that OpenAPI-based structural validation helps prevent interoperability failures, while Graph-distance algorithms and API Usage Graphs (AUGs) effectively capture semantic drifts that conventional “diff” methods often miss.

In addition, the integration of an Artificial Immune System (AIS) with sandboxed taint-assisted analysis helps detect API misuse and emerging security threats introduced during evolution. Commit-Level Filtering ensures scalability within modern DevOps pipelines, reducing computational cost by nearly 90 percent. These results demonstrate that an AI-driven, multi-dimensional approach is essential for maintaining the

integrity and interoperability of complex software ecosystems during API evolution.

B. Future Work

Although the existing framework has a solid base towards stable and safe evolution, there are a number of directions still open to future exploration:

- 1) **Automated Patch Generation:** Future versions might leverage Large Language Models (LLMs) to not only it detects semantic drift and also recommends and generates client developer code patches that are automatically generated the found fix patterns of historical commits.
- 2) **Cross-Language Analysis:** The semantic miner currently operates in a homogeneous language setting. Expanding the graph-mining capabilities to support cross-language API dependencies (e.g., a Python client consuming a Java-based microservice) would enhance the applicability of the framework in polyglot architectures.
- 3) **Collaborative Real-Time Governance:** We will also strive to create a collaborative feedback process in which the analyzer connects with developer communication systems (such as Slack/ GitHub Discussions) with NLP synthesizing community sentiment and developer “pain points” go directly to the impact score.
- 4) **Hardware-Assisted Mining:** API usage drops as API usage graphs as they become more complicated, we intend to investigate the application of special AI hardware (including TPUs or Graph Processing Units) to even lower the latency of Maximum Computations of Common Subgraph (MCS) in real-time CI/CD execution.

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